

Civics Group	Index Number	Name (use BLOCK LETTERS)	H1
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**ST. ANDREW'S JUNIOR COLLEGE
2021 JC2 PRELIMINARY EXAMINATIONS**

H1 BIOLOGY

8876/01

Paper 1: Multiple Choice

Friday

17th September 2021

1 hour

Additional Materials: Multiple Choice Answer Sheet

READ THESE INSTRUCTIONS FIRST

Do not open this booklet until you are told to do so.

Write your name, civics group and index number on the multiple choice answer sheet in the spaces provided.

There are **30** questions in this paper. Answer all questions. For each question, there are four possible answers, A, B, C and D.

Choose the one you consider correct and record your choice in soft pencil on the separate multiple choice Optical answer sheet.

INFORMATION TO CANDIDATES

Each correct answer will score one mark. A mark will not be deducted for wrong answer. Any rough working should be done in this booklet.

At the end of the examination, submit both question paper and multiple choice answer sheet.

This document consists of **19** printed pages

[Turn over

- 1** Lysosomes vary in shape and size, making them difficult to identify.

What describes a lysosome?

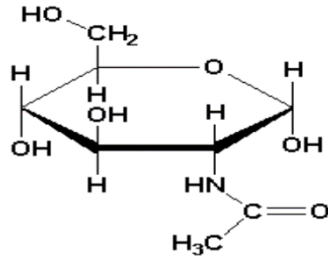
- A** a vesicle containing enzymes, enclosed by a double membrane, that is budded off the endoplasmic reticulum
- B** a vesicle containing hydrolytic enzymes and surrounded by a single membrane, found only in phagocytes
- C** a vesicle enclosed by a single membrane, containing several different hydrolytic enzymes that may act inside the cell
- D** a vesicle surrounded by a double membrane, containing enzymes which can hydrolyse damaged organelles in a cell

- 2** Which statements about prokaryotes, mitochondria or chloroplasts are correct?

- 1 Prokaryotes and mitochondria have 70S ribosomes that are the sites for translation and polypeptide synthesis.
- 2 Prokaryotes and mitochondria have an outer membrane and an inner, folded membrane where ATP synthesis occurs.
- 3 Prokaryotes and chloroplasts have circular DNA where genes coding for information are located.
- 4 Mitochondria and chloroplasts have a fully permeable inner membrane and a partially permeable outer membrane.

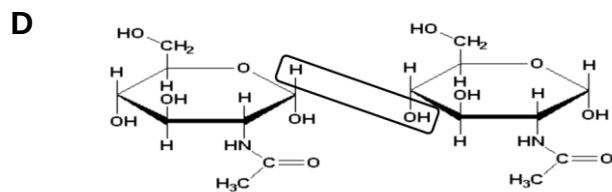
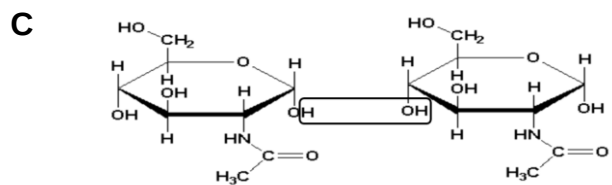
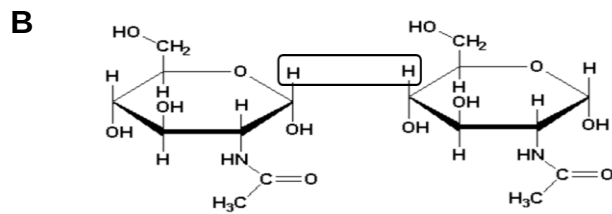
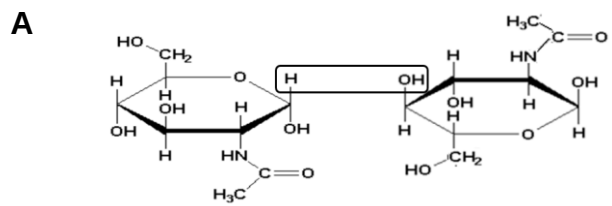
- A** 1, 2 and 3 **B** 1, 2 and 4 **C** 1 and 3 only **D** 2 and 4 only

- 3 The diagram shows the structure of a monomer N-acetyl glucosamine.



Many N-acetyl glucosamine monomers can be joined by glycosidic bonds to form a polysaccharide called chitin.

Which diagram shows the formation of a glycosidic bond between two N-acetyl glucosamine monomers?



4 Which statements about the fluid mosaic model of a membrane are correct?

- 1 The movement of phospholipids in the membrane is reduced if the fatty acid tails are saturated.
- 2 Glycoproteins in the outer layer of the membrane can move.
- 3 Channel proteins are fixed in position.

A 1, 2 and 3 **B** 1 and 2 only **C** 1 and 3 only **D** 2 and 3 only

5 The cell surface membranes of some cells are largely made up of phospholipids and cholesterol, with a lower number of transmembrane proteins than usual.

Which transport mechanisms will occur less across these membranes?

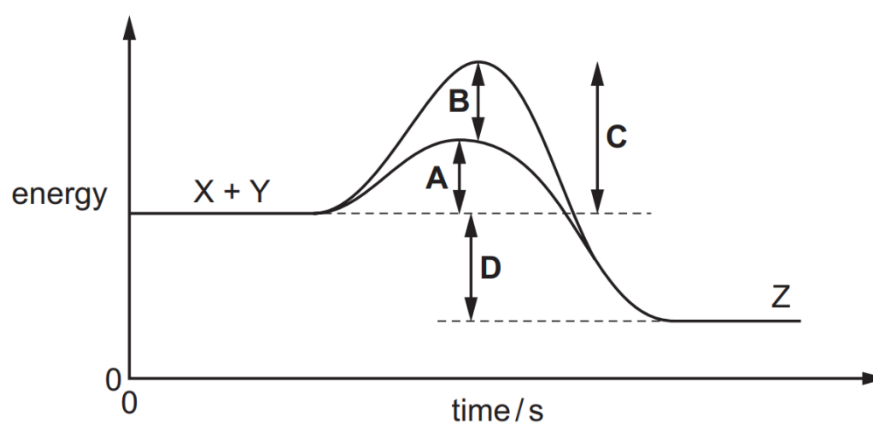
- 1 facilitated diffusion
- 2 active transport
- 3 diffusion

A 1, 2 and 3
B 1 and 2 only
C 1 and 3 only
D 2 and 3 only

6 Which row about the structure of proteins is correct?

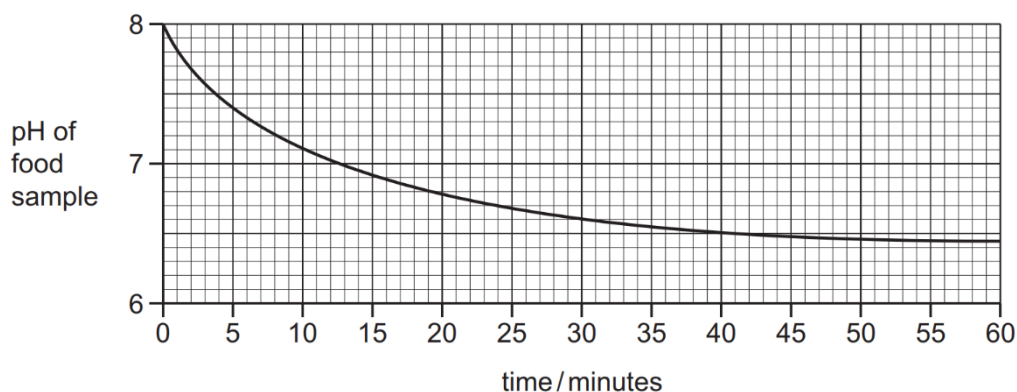
	primary structure	tertiary structure	quaternary structure
A	is the number of amino acids present in a protein	is the result of cross bonding between all the amino acids in the primary structure	is the polypeptides that link together to form a protein
B	is the order of amino acids present in a protein encoded by DNA	is the shape formed by folding of a polypeptide and held together by hydrogen bonds	contains two types of polypeptide that interact forming the shape of a protein
C	is the result of translation of an mRNA molecule by a ribosome into a chain of amino acids	is the result of ionic and hydrogen bonds, disulfide bridges and hydrophobic interactions between amino acids	is formed by four polypeptides and an additional reactive group attached to the protein
D	is the sequence of amino acids in a protein coded by an mRNA molecule	is formed as a result of interaction of the side chains of amino acids in the primary structure	is formed by the linking together of more than one polypeptide to form a protein

7 The graph shows the energy levels involved in an enzyme-catalysed reaction. Substrate molecules X and Y combine to give product Z. Which arrow shows the reduction in activation energy due to the enzyme?



- 8 Lipase is an enzyme that catalyses the hydrolysis of lipids. An experiment was carried out to investigate changes in pH when lipase is added to a food sample containing a high proportion of lipids.

The results are shown in the graph.



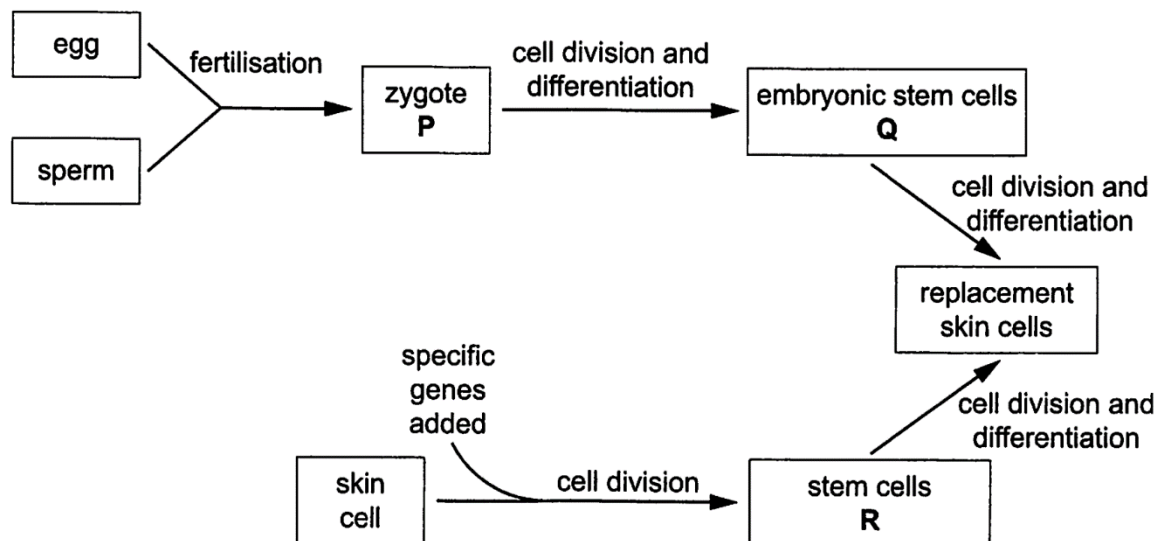
Which statements are possible explanations of the results of the experiment between 50 minutes and 60 minutes?

- 1 Enzyme concentration becomes the limiting factor.
- 2 Substrate concentration becomes the limiting factor.
- 3 All the enzyme active sites are occupied.
- 4 Denaturation of the enzyme by the products takes place.
- 5 Products formed are inhibiting the enzyme-catalysed reaction.

A 1, 2 and 3 **B** 1, 4 and 5 **C** 2, 3 and 4 **D** 2, 4 and 5

- 9 Stem cells can be used to replace cells in damaged tissues, such as skin.

The diagram shows two ways in which replacement skin cells can be produced.



Which row correctly describes the stem cells **P**, **Q** and **R**?

	P	Q	R
A	pluripotent	multipotent	totipotent
B	pluripotent	induced pluripotent	induced pluripotent
C	totipotent	pluripotent	induced pluripotent
D	totipotent	totipotent	totipotent

- 10 Nucleic acids in living cells may be single stranded or double stranded and may form hydrogen bonds between pairs of complementary bases.

Which description is correct?

- A** DNA is double stranded and, when unzipped, can form complementary base pairs with RNA.
- B** mRNA is single stranded and can form complementary base pairs only with tRNA.
- C** rRNA is single stranded and can form complementary base pairs with mRNA and tRNA.
- D** tRNA is partly double stranded and can form complementary base pairs only with mRNA.

- 11** Four nucleotides, A, B, C and D, each consist of three phosphate groups, a nitrogenous base and a pentose sugar. Characteristics of the base and sugar components before they are joined to form each nucleotide are shown in the table.

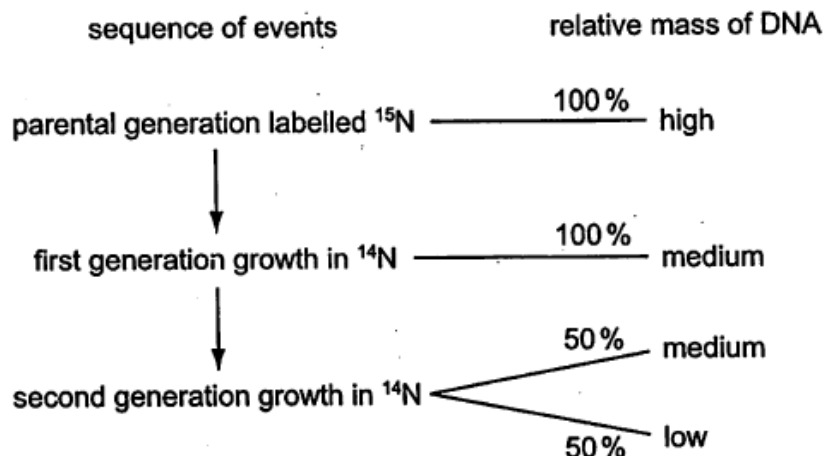
Which nucleotide could pair with an adenine base during DNA replication?

	ring structure of nitrogenous base	ratio of carbon to oxygen atoms in pentose sugar
A	double	1:1
B	double	5:4
C	single	1:1
D	single	5:4

- 12** Meselsohn and Stahl investigated DNA replication by growing bacteria in culture containing heavy nitrogen, ^{15}N , until all the DNA was labelled.

These bacteria, the parental generation, were then transferred to a medium containing only light nitrogen, ^{14}N and allowed to replicate for two generations.

DNA was extracted from each generation of bacteria and its relative mass estimated. The flow diagram shows the result.



Which row explains the mass of the DNA in the second generation grown in ^{14}N ?

	50% DNA molecules		50% DNA molecules	
	strand 1	strand 2	strand 1	strand 2
A	only ^{15}N	only ^{14}N	only ^{14}N	only ^{14}N
B	only ^{15}N	only ^{15}N	only ^{14}N	only ^{14}N
C	only ^{15}N	only ^{15}N	only ^{14}N	only ^{15}N
D	only ^{15}N	half ^{14}N , half ^{15}N	only ^{14}N	half ^{14}N , half ^{15}N

13 Some of the events that occur during transcription are listed.

- 1 Bonds break between complementary bases.
- 2 Bonds form between complementary bases.
- 3 Sugar-phosphate bonds form.
- 4 Free nucleotides pair with complementary nucleotides.

Before the mRNA molecule leaves the nucleus, which events occur twice during transcription?

- A** 1, 2 and 3 **B** 1, 3 and 4 **C** 2, 3 and 4 **D** 1 and 2 only

14 The statements describe the process of translation.

- 1 A peptide bond forms between adjacent amino acids.
- 2 Hydrogen bonds form between the anticodon and the codon.
- 3 mRNA binds to the ribosome.
- 4 tRNA enters the ribosome carrying a specific amino acid.

In which order does this process take place?

- A** 3 → 2 → 1 → 4
B 3 → 4 → 2 → 1
C 4 → 2 → 1 → 3
D 4 → 2 → 3 → 1

15 Two students were discussing the involvement of DNA and RNA in transcription and translation.

- Student 1 always stated correct facts.
- Student 2 gave further information, which was sometimes correct.

	correct facts given by student 1	further information given by student 2
1	A length of mRNA is 747 nucleotides long, including stop and start codons.	This mRNA can produce a polypeptide that is 249 amino acids long.
2	Adjacent mRNA codons of AAU and CUG bind to complementary tRNA anticodons.	There is a total of 14 hydrogen bonds formed between these two codons and their anticodons.
3	RNA polymerase catalyses the formation of the mRNA from the template strand of DNA.	During translation, an RNA adenine nucleotide will pair with a DNA thymine nucleotide.
4	A DNA adenine nucleotide is structurally different to an RNA adenine nucleotide.	The difference is in the hexose sugars; DNA is deoxyribose and RNA is ribose.

Which further information, given by student 2, is correct?

- A** 1 and 4 **B** 2 and 3 **C** 2 only **D** 4 only

16 Which description of a eukaryotic gene is correct?

- A** a sequence of bases that forms part of a DNA molecule and codes for a protein
- B** a sequence of nucleotides that forms part of a DNA molecule and codes for a polypeptide
- C** a sequence of bases that forms part of an RNA molecule and codes for a protein
- D** a sequence of nucleotides that forms part of an RNA molecule and codes for a polypeptide

- 17** The diagram shows part of the DNA sequence of a gene and a mutated sequence of the same gene.

normal DNA sequence	...CCG GAT TAT TGC GAG AAA TGG CAT TCT AGG...
mutated DNA sequence	...CCG GAT GTA TTG CGA GAA ATG CAT TCT AGG...

What are possible effects of the mutated sequence?

- 1 the presence of mRNA stop codons, UAG, UAA or UGA
- 2 a change in the sequence of amino acids
- 3 a non-functional protein
- 4 ribosomes cannot translate the mRNA

- A** 1, 2 and 3
B 1, 3 and 4
C 1 and 4 only
D 2 and 3 only

- 18** Down's syndrome results from chromosomal aberration of chromosome 21.

This can occur in several ways.

- 1 During gamete formation the homologous pair of chromosomes of chromosome 21 fail to separate. A gamete with two copies of chromosome 21 is fertilised by a gamete with one copy of chromosome 21.
- 2 During early development of the embryo, chromatids of a chromosome 21 fail to separate some cells.
- 3 Translocation of part of chromosome 21 onto chromosome 14 during the development of the embryo. During gamete formation, chromosomes separate normally but may contain an extra chromosome 21.

Some individuals have a form of Down's syndrome called mosaicism. In these cases some of the body cell have the usual two copies of chromosome 21 and other cells have three copies of chromosome 21.

Which of the ways could explain how the mosaicism form of Down's syndrome occurs?

- A** 1 and 2
B 1 and 3
C 2 only
D 3 only

19 At which stages of mitosis are chromosomes composed of two chromatids that are held together by a centromere?

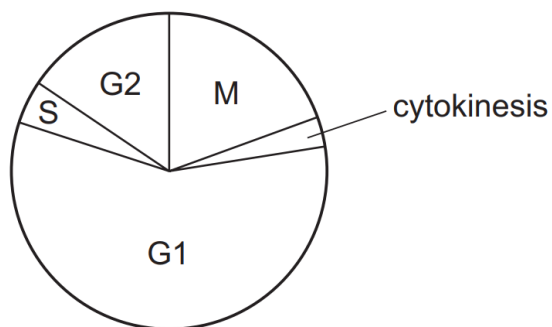
- A** anaphase and telophase
- B** anaphase and prophase
- C** metaphase and prophase
- D** metaphase and telophase

20 Which statements about mitosis are correct?

- 1 Mitosis results in genetically identical cells.
- 2 Mitosis is involved in asexual reproduction.
- 3 Mitosis occurs immediately after cytokinesis.
- 4 Mitosis can be used to repair damaged cells.

- A** 1, 2, 3 and 4
- B** 1, 2 and 4 only
- C** 1 and 2 only
- D** 3 and 4 only

21 The diagram shows the cell cycle of a mammalian cell.



Cell cycle regulation was done to prevent the cycle from continuing when mistakes are made or when DNA is damaged.

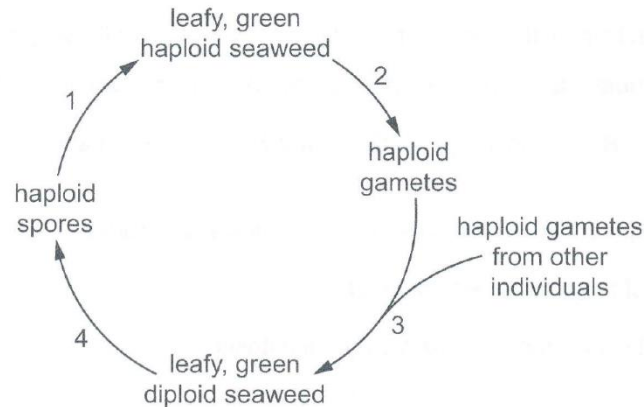
Four of the processes to regulate the cell cycle are described.

- 1 Mitosis is blocked if DNA replication is incomplete.
- 2 Anaphase is blocked if the assembly of chromatids on the spindle is unsuccessful.
- 3 DNA replication is blocked if DNA is damaged.
- 4 DNA replication stops if damage to DNA has not been repaired.

In which phases of the cell cycle do these processes occur?

	Phases			
	1	2	3	4
A	M	G1	S	G2
B	G2	M	G1	S
C	G2	S	G1	M
D	S	G2	M	G1

- 22 The life-cycle of a seaweed has a haploid stage and a diploid stage.



At which of the steps in this life-cycle will genetic variation be introduced?

- A 1 and 2
 - B 2 and 3
 - C 3 and 4
 - D 4 only
- 23 Which statement uses genetic terms correctly?
- A For a gene with multiple alleles, whether each allele is dominant or recessive is independent of the other alleles.
 - B For a gene with two alleles showing incomplete dominance, the genotypic ratio in a cross involving two heterozygotes is always the same as the phenotypic ratio.
 - C Homologous chromosomes have the same alleles at the same loci.
 - D X-linked recessive diseases only occur in males since females can only be carriers.
- 24 In humans, a certain disease is caused by inheriting two recessive alleles of gene A/a.

Identical twins occur once in approximately 300 births.

A woman who is heterozygous for gene A/a is about to give birth. The father is also heterozygous for this gene.

What is the probability that the woman will give birth to identical twins that do **not** have the disease?

- A approximately 1 in 300
- B approximately 1 in 400
- C approximately 1 in 800
- D approximately 1 in 1200

- 25** An unspotted black female dog was crossed with a spotted black male dog. She gave birth to six puppies of four different phenotypes: unspotted black, unspotted brown, spotted black and spotted brown. Some of the puppies were adopted and live in different places. As the puppies got older, these phenotypes did not change.

Some of the phenotypes of the puppies were different from the phenotypes of either of their parents.

What is needed to explain the appearance of these new phenotypes?

- A** Codominance and meiosis
 - B** Multiple alleles and random fertilisation
 - C** Environmental factors and random fertilisation
 - D** Random fertilisation and meiosis
- 26** Different stages in aerobic respiration of one molecule of glucose are associated with processes in particular regions of cells.

Molecules of ATP, reduced NAD and reduced FAD are produced as a result of these different processes.

- 1 2 reduced NAD
- 2 2 reduced NAD and 2 ATP
- 3 6 reduced NAD, 2 reduced FAD and 2 ATP
- 4 26-28 ATP

What is the contribution to the net gain from aerobic respiration of one molecule of glucose, as a result of the processes in different regions of the cell?

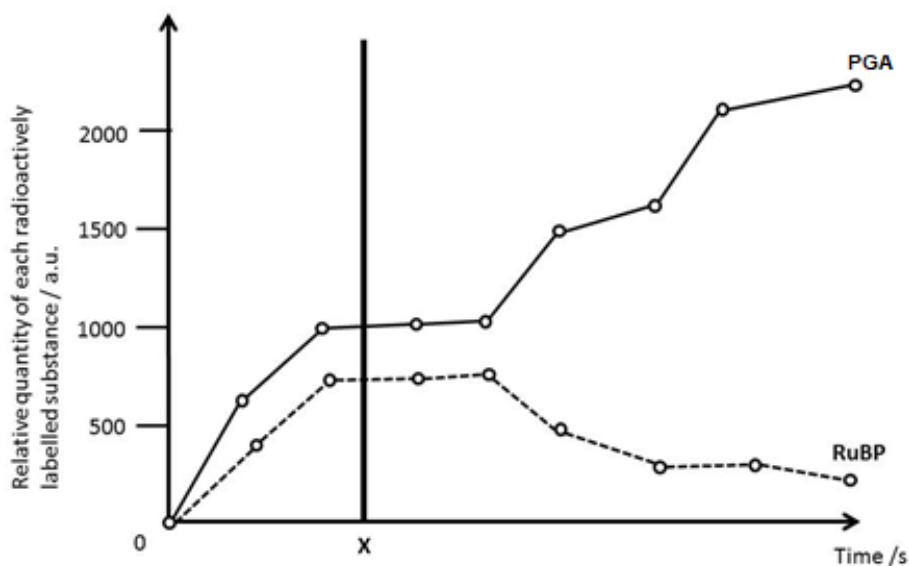
	cytoplasm	inner membrane of mitochondrion	passage through both mitochondrial membranes	mitochondrial matrix
A	1	2	3	4
B	2	4	1	3
C	3	1	4	2
D	4	3	2	1

- 27 Which of the following statement about anaerobic respiration is correct?
- A Animal cells produce pyruvate from lactate, so as to regenerate NAD^+ .
 - B Plant cells use their chloroplasts to synthesise ATP for cellular activities when aerobic respiration in mitochondria cannot occur.
 - C From one molecule of glucose, aerobic respiration yields 10 to 15 times more ATP than anaerobic respiration.
 - D From one molecule of glucose, anaerobic respiration in both plant and animal cells will yield the same amount of ATP.

- 28** Many chemicals are used as herbicides as they are able to inhibit photosynthesis in plants. Diquat, a herbicide, inhibits Photosystem I by intercepting electrons from an electron carrier.

To investigate the effect of Diquat on light-independent reactions, an illuminated suspension of photosynthesizing algae *Chlorella* was treated with carbon dioxide containing a heavy isotope of carbon, ^{14}C . The levels of ribulose biphosphate (RuBP) and glycerate-3-phosphate (PGA) were subsequently measured.

The figure shows the results of the experiment. Point X indicates the point at which Diquat was added to the *Chlorella* suspension.



Which row correctly explains why the level of PGA increases and RuBP decreases after point X?

	<i>Carbon fixation</i>	<i>PGA reduction</i>	<i>RuBP regeneration</i>
A	✓	✓	✗
B	✗	✗	✓
C	✓	✗	✗
D	✓	✗	✓

- 29 In which of the following situations would evolution be slowest for an interbreeding population?

	Migration	Selection pressure	Variation due to mutation
A	Absent	Low	Low
B	Absent	High	High
C	High	Low	High
D	High	High	Low

- 30 Soay sheep, a primitive breed of domesticated sheep, live wild on the small and remote Scottish island of Hirta. Studies have shown that the mean size of an adult Soay sheep has been decreasing over the past thirty years.

Which statements could explain the decrease in mean size of adult Soay sheep on Hirta?

- 1 Stabilising selection is occurring, with the largest and smallest sheep being selected against.
- 2 Small lambs are less likely to survive their first winter than large lambs.
- 3 Weather change has led to milder winters, so smaller lambs are surviving to adulthood.
- 4 Food has become scarcer and smaller sheep need less food than larger sheep.

- A** 1 and 3
B 1 only
C 2 and 4
D 3 and 4

END OF PAPER

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ANSWERS:

Q	Ans	Q	Ans
1	C	16	B
2	C	17	D
3	C	18	C
4	B	19	C
5	B	20	C
6	D	21	B
7	B	22	C
8	D	23	B
9	C	24	B
10	A	25	D
11	D	26	B
12	A	27	D
13	D	28	C
14	B	29	A
15	C	30	D